Assignment 6

1a.) $x + 3 \bmod_{26}$ $\rightarrow$ (MA TE MAHI KA ORA)

b.i) $e(x) = 7x + 10$

ii $d(y) = 15y + 6$ as $(y-10) \cdot 7^{-1} \bmod_{26}$

2a.) As, $p = 19$, $q = 29$ $\phi(n) = 18 \cdot 28 = 504$
the $e^{-1} \bmod \phi$ = gcd$(295, 504) = 1$ $(295 = 5 \times 59)$ & $(504 = 2^3 \times 3^2 \times 7)$ $\therefore$ there are no prime factors $\because$ gcd$(295, 504) = 1$

b.) As $504(24) - 295(41) = 1$, that $-295 \cdot 41 \equiv 1 \bmod_{504}$
As $-41 \equiv 463 \bmod_{504}$, $d = 463$

c.) if B, $(32)^{295} \bmod_{551} \equiv (2^5)^{295} \bmod_{551}$

$\therefore (2^5)^{295} \bmod_{551} \equiv 200 \bmod_{551}$ $(2^{504} \equiv 1 \bmod_{551})$

3.) Alice & Bob use $n = 61771$, $e = 5$

$n = pq = 61771$ $(p-1)(q-1) = 61272$

$\therefore p + q = 500$ for $x^2 - 500x + 61771$

$$\frac{500 \pm \sqrt{500^2 - 4 \cdot 61771}}{2} = \frac{500 \pm 54}{2} = 277 \,\&\, 223$$